HW4: Problem-2 - RSA Implementation via SageMath

Overview

RSA (Rivest–Shamir–Adleman) is a widely used public-key cryptosystem. Its security relies on the difficulty of factoring the product of two large prime numbers. For this task, a 256-bit modulus (n) was generated using two 128-bit primes to demonstrate the core operations of Key Generation, Encryption, and Digital Signatures.

SageMath Implementation

The implementation was performed using the `SageMath Cell` server. The script generates prime numbers, calculates the private and public exponents, and performs modular exponentiation for encryption and signing.

The following screenshots show the SageMath environment, including the source code and the resulting hexadecimal output.

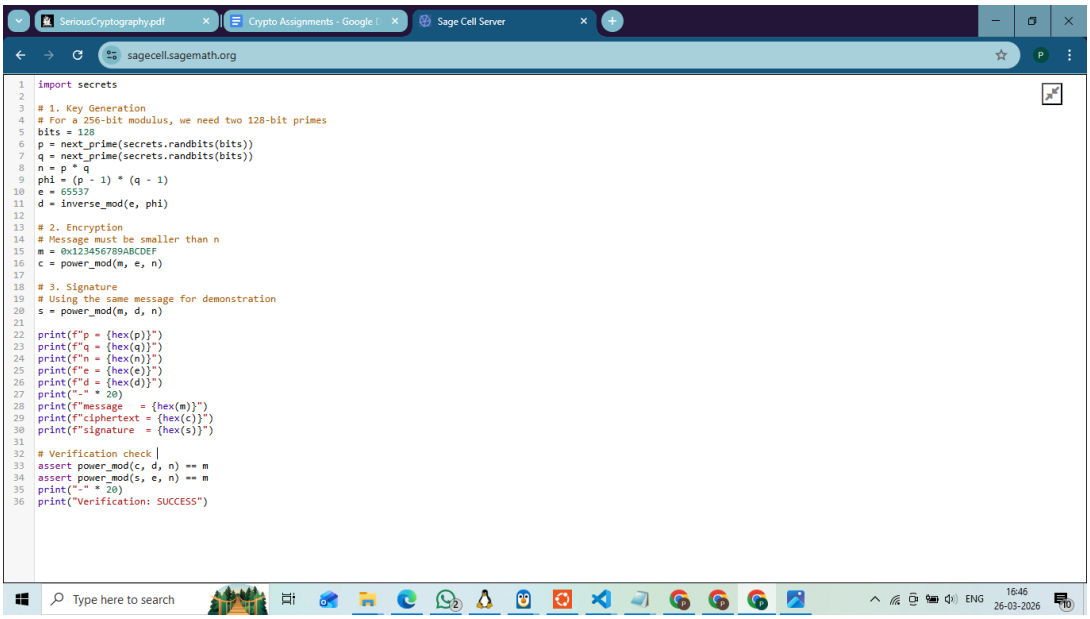

Figure 1: SageMath script for RSA Key Generation and Operations.

Figure 2: Formatted hex output and successful verification.